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Roll No.:

**Third SEMESTER-B.TECH.
END-SEMESTER THEORY EXAMINATION,
Nov-Dec 2024**

Course Code-CMMTC303

Course Title-Applied Linear Algebra

Time: 3:00 hours

Max Marks-50

Note: Attempt all the five questions. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q.1.	Attempt any 2 parts of the following		
1(a)	Show that the set A is a vector space over $\mathbb{R}$, where $A = \{(x, y, z) \mid x + y + 2z = 0 = 2x + y + z\}$.	5	CO1
1(b)	Find the bases of row, column and null spaces for the matrix $A = \begin{bmatrix} 1 & 2 & 0 & 2 & 5 \\ -2 & -5 & 1 & -1 & -8 \\ 0 & -3 & 3 & 4 & 1 \\ 3 & 6 & 0 & -7 & 2 \end{bmatrix}$.	5	CO1
1(c)	Find the basis and the dimension of the subspaces $U + V$ and $U \cap V$ of $\mathbb{R}^5$, where $U = \text{span}\{(1, 3, -2, 2, 3), (1, 4, -3, 4, 2), (2, 3, -1, -2, 9)\}$ and $V = \text{span}\{(1, 3, 0, 2, 1), (1, 5, -6, 6, 3), (2, 5, 3, 2, 1)\}$.	5	CO1
Q.2.	Attempt any 2 parts of the following		
2(a)	Verify rank-nullity theorem for a linear transformation $T : \mathbb{R}^5 \rightarrow \mathbb{R}^3$ such that $T(x, y, z, s, t) = (x + 2y + 2z + s + t, x + 2y + 3z + 2s - t, 3x + 6y + 8z + 5s - t)$.	5	CO2
2(b)	Let $T : P_3(\mathbb{R}) \rightarrow P_3(\mathbb{R})$ be a linear transformation such that $T(f(x)) = f''(x) - 4f'(x) + f(x)$. Find the associated matrix $[T]_\beta$ for the basis $\beta = \{x, 1 + x, x + x^2, x^3\}$.	5	CO2
2(c)	Show that $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is an invertible linear transformation such that $T(x, y, z) = (2x, 4x - y, 2x + 3y - z)$ and find the formulas for T^2 and T^{-1} .	5	CO2
Q.3.	Attempt any 2 parts of the following		
3(a)	Classify all possible Jordan canonical forms of a 5×5 matrix A that has a single eigenvalue λ of algebraic multiplicity 5.	5	CO3

3(b)	Diagonalize the matrix $A = \begin{bmatrix} 5 & -4 & 4 \\ 12 & -11 & 12 \\ 4 & -4 & 5 \end{bmatrix}$ and also evaluate $\ A\ _1, \ A\ _2$ and $\ A\ _\infty$.	5	CO3
3(c)	Use Cayley-Hamilton theorem to find the inverse of the matrix $A = \begin{bmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix}$ and hence deduce the value of $A^5 - 6A^4 + 6A^3 - 11A^2 + 2A + 3I$.	5	CO3
Q.4.	Attempt any 2 parts of the following		
4(a)	Consider the space $P_2(\mathbb{R})$ endowed with the inner product $\langle f(x), g(x) \rangle = \int_{-1}^1 f(t)g(t)dt$. Find its orthonormal basis using Gram Schmidt process and also find orthogonal projection of $h(x) = x + 1$ on $P_2(\mathbb{R})$.	5	CO4
4(b)	Let V be a finite dimensional inner product space over the field F and let $\beta = \{v_1, v_2, \dots, v_n\}$ be an orthonormal subset of V . Then for all $x, y \in V$; prove the following: (i) $ \langle x, y \rangle \leq \ x\ \cdot \ y\ $. (ii) $\ x\ ^2 \geq \sum_{i=1}^n \langle x, v_i \rangle ^2$.	5	CO4
4(c)	Let $T : V \rightarrow V$, where V is a finite dimensional inner product space over some field F . Prove the following: (i) Every eigenvalue of T is real, if T is a self-adjoint operator. (ii) $T - \lambda I$ is normal, if T is a normal operator, where $\lambda \in F$. (iii) $\langle T(x), T(y) \rangle = \langle x, y \rangle \forall x, y \in V$, if T is a unitary operator.	5	CO4
Q.5.	Attempt any 2 parts of the following		
5(a)	Verify that the function $H : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}$ is a bilinear form, where $H((a_1, a_2, a_3), (b_1, b_2, b_3)) = a_1b_1 - 2a_1b_2 + a_2b_1 - a_3b_3$ and compute its matrix representation with respect to the basis $\beta = \{(1, 0, 1), (1, 0, -1), (0, 1, 0)\}$.	5	CO5
5(b)	Find the rank, signature and index for the matrices representing each of the following quadratic forms: (i) $x_1^2 + 4x_1x_2 + 3x_2^2$, (ii) $x_1^2 - 2x_2^2 - 3x_3^2 + 4x_1x_2 + 6x_1x_3 - 8x_2x_3$.	5	CO5
5(c)	Determine the critical point and its nature for the following functions: (i) $f(x, y) = 3x^2 - xy + y^2$. (ii) $f(x, y, z) = x^3 + xyz + y^2 - 3z$.	5	CO5

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